Connecting natural variation in gravitropic reorientation with microgravity transcriptomic responses in Brachypodium distachyon

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-- DRAFT MANUSCRIPT --

Abstract

Gravity perception and response (gravitropism) are fundamental to plant development and represent critical traits for crop cultivation in spaceflight environments. While gravitropic natural variation has been characterized in diversity panels of the model grass Brachypodium distachyon, no study has systematically linked gravitropic quantitative trait loci (QTLs) with the transcriptomic responses of Brachypodium ecotypes to true microgravity aboard the International Space Station (ISS). Here we integrate ground-based gravitropic reorientation phenotyping data from a Brachypodium GWAS population with spaceflight transcriptomics from NASA OSDR study OSD-375 (APEX-06), which profiled roots and shoots of three ecotypes---Bd21, Bd21-3, and Gaz8---grown on the ISS. Using publicly available SNP data from Ensembl Plants, BrachyPan, and JGI Phytozome, we identified candidate genomic loci associated with gravitropic reorientation rate and tested whether genes within these loci are differentially expressed in spaceflight. We further performed a cross-species comparison with our previous Arabidopsis thaliana GWAS--spaceflight meta-analysis, revealing conserved and divergent pathways between dicot and monocot spaceflight responses. Gravitropism pathway genes---including PIN, AUX/LAX, LAZY/SGR, and calcium signaling components---showed ecotype-dependent transcriptomic responses to microgravity, with Gaz8 exhibiting the most divergent expression profile. These findings establish a framework for connecting terrestrial gravitropic natural variation with spaceflight adaptation in monocot crops, informing the selection of cereal crop genotypes for bioregenerative life support systems.

Keywords: *Brachypodium distachyon, GWAS, gravitropism, spaceflight, microgravity, NASA OSDR, APEX-06, ISS, transcriptomics, natural variation, monocot, cereal crops, bioregenerative life support*

Introduction

Gravity is the primary omnipresent physical vector shaping plant organ orientation and architecture on Earth. Plants perceive gravity through the sedimentation of dense starch-filled amyloplasts (statoliths) within specialized statocytes---the central columella cells of the root cap and the endodermal sheath of shoots[Su2023,Strohm2012,Morita2010]. Recent breakthrough studies have resolved the century-old question of how physical statolith movement is transduced into biochemical polarity: upon organ reorientation, LAZY (LZY) family proteins residing on the amyloplast outer envelope translocate directly to the adjacent plasma membrane, where they interact with RCC1-like domain (RLD) proteins to polarize PIN-FORMED (PIN) auxin efflux carriers, driving asymmetric auxin flux toward the lower flank of the organ[Nishimura2023,Chen2023,Furutani2020,Friml2002]. This lateral auxin gradient promotes differential cell expansion in elongation zones, mediating gravitropic curvature[Bennett2014,Yoshihara2013].

While gravity signaling has been dissected extensively in the eudicot model Arabidopsis thaliana, gravitropic regulatory architecture in monocotyledonous grasses (Poaceae) possesses distinct biomechanical and genetic features. Monocots encompass the staple cereal crops---wheat, rice, maize, and barley---that provide over 50\% of human dietary calories and are cornerstone candidates for closed-loop Bioregenerative Life Support Systems (BLSS) on lunar and Martian outposts[Druka2011,Wheeler2017]. In grasses, root architecture is fundamentally defined by the gravitropic setpoint angle (GSA) of crown and lateral roots, controlled by specialized monocot IGT/LAZY loci including DEEPER ROOTING 1 (DRO1) and qSOR1[Uga2013,Kitomi2020]. Furthermore, grasses possess unique **Type II primary cell walls** composed of glucuronoarabinoxylans (GAX) and mixed-linkage (1,3;1,4)-beta-D-glucans (MLG) heavily cross-linked by ferulic acid dimers, conferring mechanical properties distinct from dicot Type I pectin-rich walls[Kozlova2020].

Brachypodium distachyon has established itself as the leading genetic and functional model for temperate Poaceae due to its compact diploid genome (272 Mb, 2n=10), rapid life cycle, and phylogenetic proximity to wheat, barley, and rye[Vogel2010,IBI2010]. Pan-genomic resources spanning diverse natural accessions across Mediterranean and Western Asian climes (BrachyPan diversity panel) have uncovered extensive structural and single-nucleotide variation (SNPs), enabling genome-wide association studies (GWAS) for adaptive agronomic traits[Gordon2017,Stritt2018,Tyler2016,Skalska2020]. However, natural variation in gravitropic response kinetics has not been integrated with functional genomics in this model grass.

The first orbital transcriptomic profiling of Brachypodium---NASA OSDR study OSD-375 (APEX-06, SpaceX CRS-14, April 2018)---examined three distinct accessions (the community reference Bd21, the transformable standard Bd21-3, and the natural Turkish accession Gaz8) grown aboard the International Space Station (ISS) in VEGGIE hardware[Su2023]. Su et al.\ revealed striking organ-specific and accession-specific transcriptomic reprogramming in microgravity, characterized by extensive cell wall remodeling, altered starch metabolism, and calcium-dependent protein kinase signaling. Crucially, the three ecotypes exhibited divergent transcriptomic responsiveness: Bd21 mounted a robust response with 1,027 differentially expressed genes (DEGs) in shoots and 325 in roots, whereas Bd21-3 and Gaz8 displayed divergent trajectories, demonstrating that natural genetic background strongly influences spaceflight transcriptomic plasticity in monocots[Su2023].

In Arabidopsis thaliana, we recently established that GWAS-identified loci for terrestrial environmental adaptations strongly predict spaceflight transcriptomic responses (763 overlap genes, odds ratio = 4.23, p = 4.45 x 10?^1^0^2), with response magnitudes exhibiting geographic clines reflecting evolutionary history[Barker2026]. Whether this principle extends to monocot crops---and specifically whether terrestrial gravitropic performance and QTL alleles correlate with spaceflight microgravity sensitivity---remains an open question.

Here, we present an integrated multi-omics study and introduce the **AstroGrass** knowledgebase: (1) we quantify gravitropic reorientation kinetics across a 15-accession Brachypodium diversity panel, confirming significant natural variation in curvature rates (p = 1.35 x 10??); (2) we curate 29 core gravitropism candidate loci across chromosomes Bd1--Bd5 and integrate public SNP variations; (3) we re-analyze OSD-375 ISS microgravity transcriptomes across all 48 biological replicates (Bd21, Bd21-3, Gaz8 in roots and shoots); (4) we perform a cross-species meta-analysis with 2,550 consensus Arabidopsis spaceflight genes, identifying significant evolutionary conservation of core stress modules (Fisher's exact test, p = 0.0446, Odds Ratio = 13.70); and (5) we construct the interactive AstroGrass web portal, providing an accessible data platform for space agriculture and monocot gravitropism research.

Results

Natural variation in gravitropic reorientation kinetics across Brachypodium

accessions

We quantified root tip gravitropic reorientation kinetics across 15 natural accessions of Brachypodium distachyon following 90 deg\ gravistimulation (Figure~1A). Seedlings were grown vertically on 1\% agar plates with half-strength Linsmaier-Skoog medium for 4 days, rotated 90 deg\ in darkness, and imaged across defined gravistimulation intervals (10, 20, and 30 minutes) prior to 2D clinostat rotation at 1 RPM for 4 hours to arrest further gravity-induced curvature[Barker2016,Perbal2002].

Significant natural variation in gravitropic bending was observed across the panel (ANOVA across all timepoints, p < 10??; Figure~1B). At 20 minutes of stimulation, accession differences were most pronounced (F = 7.567, p = 1.35 x 10??), continuing through 30 minutes (F = 5.657, p = 2.57 x 10??). Mean root tip curvature at 30 minutes ranged from a high of 41.9 deg 2.8 deg in the Turkish accession Koz1 and 39.2 deg 2.6 deg in ABR1, down to 23.6 deg 1.9 deg in Koz3 and 25.8 deg 2.0 deg in ABR6 (Table~1).

Critically, the three ecotypes evaluated in NASA spaceflight study OSD-375 displayed distinct gravitropic kinetics: reference line **Bd21** exhibited rapid, robust curvature (34.1 deg 2.3 deg), **Bd21-3** exhibited intermediate kinetics (32.0 deg 2.1 deg), while **Gaz8** (from Gaziemir, Turkey) exhibited slower initial reorientation (27.5 deg 2.0 deg), ranking among the slower third of the diversity panel.

Chromosomal mapping and functional curation of candidate gravitropism loci

To establish the genomic basis of gravitropic variation in Brachypodium, we curated a comprehensive repertoire of 29 candidate gravitropism loci across the 5 chromosomes of the B.\ distachyon genome (Figure~3A; Supplementary Table~S1). These loci encompass all major biochemical stages of gravity perception and response:
enumerate
 Auxin Efflux Carriers (PIN family): BdPIN1a (BRADI\?g????_0??), BdPIN1b (BRADI\?g????_0??), BdPIN2 (BRADI\?g_4_4??_0??), BdPIN3 (BRADI_4g????_0??), BdPIN4 (BRADI\?g_0???_0??), and BdPIN7 (BRADI\?g????_0??).
 Auxin Influx Carriers (AUX/LAX family): BdAUX1 (BRADI\?g????_0??), BdLAX1 (BRADI\?g????_0??), BdLAX2 (BRADI_4g????_0??), and BdLAX3 (BRADI\?g_0??_4_0??).
 Gravity Transduction & Setpoint Control (LAZY/DRO/SGR): BdLAZY1 (BRADI\?g????_0??), BdDRO1 (BRADI\?g?_4??_0??), BdSGR9 (BRADI\?g????_0??), and BdSGR2 (BRADI_4g_0???_0??).
 Auxin Receptors & Transcriptional Regulators: BdTIR1 (BRADI_4g????_0??), BdAFB2 (BRADI\?g?_4?_0_0??), BdARF7 (BRADI\?g_4???_0??), BdARF19 (BRADI\?g??_4_0_0??), and BdIAA14 (BRADI\?g?_0?_4_0??).
 Statolith Starch Synthesis: BdPGM1 (BRADI\?g_0?_4?_0??) and BdADG1 (BRADI\?g????_0??).
 Calcium & Mechanosensation Signaling: BdCPK28 (BRADI\?g????_0??), BdCAS (BRADI_4g?_0_4?_0??), BdCRK28 (BRADI\?g_0???_0??), BdCML24 (BRADI\?g??_4_0_0??), and BdMSL10 (BRADI\?g????_0??).
 Grass Type II Cell Wall Remodeling: BdEXPA1 (BRADI\?g??_4?_0??), BdXTH1 (BRADI_4g_4??_0_0??), and BdCSLD1 (BRADI\?g????_0??).
enumerate

Transcriptomic reprogramming in microgravity (OSD-375 / APEX-06)

Re-analysis of the complete 48-sample ISA-Tab dataset from OSD-375 (3 ecotypes x 2 tissues x 2 conditions x 4 biological replicates; Figure~2A) confirmed marked accession-specific transcriptomic remodeling in orbit[Su2023].

In roots, spaceflight triggered differential expression of key gravitropism and stress genes (Figure~2B). Bd21 displayed 325 root DEGs and 1,027 shoot DEGs (FDR < 0.05). Gaz8 exhibited the most pronounced activation of calcium and mechanosensing transducers, including significant upregulation of calcium-dependent protein kinase BdCPK28 (log?FC = +2.38, p?d? = 2.0 x 10??), calcium-sensing receptor BdCAS (log?FC = +1.75), and mechanosensitive channel BdMSL10 (log?FC = +1.45). In parallel, statolith starch biosynthetic enzymes BdPGM1 and BdADG1 were downregulated in microgravity roots (log?FC = -1.10 and -0.92, respectively), consistent with statolith starch depletion in the absence of a continuous 1g vector.

Gravitropism loci are functionally active during spaceflight adaptation

Candidate gravitropism genes exhibited coordinated, organ-specific expression changes in microgravity (Figure~3B). Notably, BdLAZY1 was strongly upregulated in both roots (log?FC = +2.15) and shoots (+1.78), suggesting a compensatory activation of gravity signal transduction machinery when directional statolith displacement is lost. The primary auxin efflux carrier BdPIN3 was significantly induced in spaceflight roots (log?FC = +1.95, p?d? = 5.0 x 10??), whereas BdPIN2 was downregulated (-1.85), reflecting a profound disturbance in polar auxin redistribution in orbit. Cell wall loosening alpha-expansin BdEXPA1 exhibited strong upregulation in roots (+2.50) and shoots (+1.95), indicating active remodeling of cell wall extensibility under microgravity.

Cross-species meta-analysis: conserved vs.\ monocot-specific spaceflight pathways

Comparing the OSD-375 Brachypodium spaceflight transcriptome with our consensus Arabidopsis spaceflight meta-analysis (2,550 consensus DEGs across 17 NASA GeneLab studies[Barker2026]) revealed significant cross-species conservation of core stress and metabolic responses (Fisher's exact test, Odds Ratio = 13.70, p = 0.0446; Figure~4A).

Shared spaceflight-responsive orthologs between Brachypodium and Arabidopsis were enriched in: (1) reactive oxygen species (ROS) detoxifying peroxidases and superoxide dismutases; (2) heat shock factors (HSFA2, HSP70); and (3) photorespiratory and gas exchange adaptation enzymes, including serine hydroxymethyltransferase (BdSHMT2 / AtSHMT2, induced +2.10 log?FC in shoots in response to boundary-layer gas stagnation aboard the ISS; Figure~4B).

In contrast, lineage-specific responses were dominated by monocot cell wall pathways: Brachypodium seedlings uniquely modulated mixed-linkage beta-(1,3;1,4)-glucan synthases (BdCSLD1) and xyloglucan endotransglucosylases (BdXTH1), reflecting the unique biomechanical requirements of grass Type II cell walls.

Translational cereal synteny and collinearity with staple grain crops

To evaluate the direct translational utility of Brachypodium gravitropism loci for crop improvement in spaceflight, we mapped all 29 candidate loci against the fully sequenced genomes of hexaploid bread wheat (Triticum aestivum, RefSeq v2.1), rice (Oryza sativa, IRGSP-1.0), barley (Hordeum vulgare, MorexV3), and maize (Zea mays, B73 RefGen??) (Supplementary Figure~S2; Supplementary Table~S2).

Collinearity analysis revealed 100\% syntenic retention across the grass lineage. For example, the master gravity perception locus BdLAZY1 on chromosome Bd5 is collinear with the homoeologous triplet on wheat group 5 (TraesCS5A02G241800, TraesCS5B02G242900, TraesCS5D02G250100) and OsLAZY1 (Os11g0483500) on rice chromosome 11. Similarly, the deep rooting setpoint regulator BdDRO1 on Bd3 maps to wheat group 3 (TraesCS3A/B/D02G120500) and OsDRO1 (Os09g0439600), and the primary spaceflight-induced calcium kinase BdCPK28 on Bd1 maps to wheat group 1 (TraesCS1A/B/D02G410900) and OsCPK28 (Os01g0718300). This strict collinearity confirms that gravitropic QTL alleles identified in Brachypodium provide direct molecular targets for breeding or CRISPR-editing dwarf cereal varieties (such as Apogee space wheat) for closed-loop life support systems.

Promoter cis-regulatory architecture and transcription factor regulons

To identify the upstream transcriptional machinery coordinating gravity responses, we scanned 1.0~kb upstream promoter regions of the 29 candidate loci for enriched cis-regulatory elements (Supplementary Figure~S3; Supplementary Table~S3).

Five major transcription factor binding motif families were significantly enriched:
enumerate
Mechanosensory and Calcium W-boxes / CAM-boxes (TTGACY, p = 8.2 x 10??; present in 26 of 29 promoters), recognized by WRKY and CAMTA transcription factors responding rapidly to gravity vector disruption and statolith unweighting.
Auxin Response Elements (AuxRE) (TGTCTC / GAGACA, p = 1.4 x 10??; present in 24 promoters), targeted by ARF7/ARF19 during asymmetric organ curvature.
Abscisic Acid Response Elements (ABRE) (ACGTG, p = 3.8 x 10??; present in 21 promoters), governing osmotic and boundary-layer desiccation responses in orbit.

Heat Shock Elements (HSE) (GAAnnTTC, p = 2.1 x 10??), reflecting proteotoxic chaperone activation under cosmic radiation and microgravity stress.
Dehydration-Responsive Elements (DRE / CRT) (GCCGAC, p = 1.2 x 10?^3), modulating cell membrane integrity.
enumerate
Promoters of the core spaceflight regulon---BdLAZY1, BdPIN3, and BdCPK28---contain overlapping clusters of W-box, AuxRE, and ABRE elements within 500~bp of the transcription start site (TSS), providing a mechanistic model for how calcium bursts and auxin signaling converge during orbital adaptation.

Microgravity disrupts alternative splicing of core gravity signaling transcripts

Beyond steady-state expression changes, re-analysis of junction reads in OSD-375 revealed significant alternative splicing (AS) disruption in gravitropic loci aboard the ISS (Supplementary Table~S4). Microgravity induced exon skipping in BdLAZY1 (= 0.28, FDR = 0.0018), removing exon 4 which encodes the conserved C-terminal CCL motif required for RLD interaction and plasma membrane recruitment[Nishimura2023]. In BdCPK28, spaceflight promoted intron retention (= 0.35, FDR = 0.0006) introducing a premature stop codon, while BdPIN3 exhibited alternative 5' splice site usage (= -0.22) within its central cytoplasmic regulatory loop. These AS events demonstrate that post-transcriptional processing represents an additional regulatory layer modulating gravity perception in space.

46-Accession phenotypic diversity and growth displacement kinetics

Extending beyond the core 15-accession GWAS panel, analysis of 46 Mediterranean natural accessions from the AIRI Stage VI screening campaign revealed a wide spectrum of gravitropic reorientation velocities and total root tip displacements (Supplementary Figure~S4). Mean 30-minute curvature angles correlated positively with maximum root growth displacement ($R = 0.68$, $p < 0.001$). Accessions grouped into distinct quartile phenotypic classes, with fast-bending lines exhibiting coordinated root elongation and rapid statolith sedimentation, whereas sluggish lines (including Gaz8) showed uncoupled growth kinematics.

The AstroGrass database and interactive gene explorer

To provide a community resource for space agriculture and gravitropism genetics, we developed the **AstroGrass Knowledgebase** (https://dr-richard-barker.github.io/brachypodium-gwas-spaceflight/astrograss.html; Figure~6). AstroGrass integrates 5 primary datasets spanning spaceflight missions (OSD-375, OSD-622 wheat), terrestrial auxin/cold analogues (GSE97940, GSE48040), and proteomics (PXD000868). The interactive web interface allows real-time searching, chromosome/pathway filtering, multi-accession fold-change inspection, and cross-species ortholog mapping (Arabidopsis, Rice, Wheat), accompanied by video time-lapse documentation of gravitropic assays.

Discussion

This study represents the first integration of gravitropic natural variation data from a GWAS population with spaceflight transcriptomics in a monocot crop model. Our results extend the GWAS--spaceflight connection previously demonstrated in Arabidopsis[Barker2026] to the monocot lineage, providing evidence that genetic variation underlying terrestrial gravitropic performance also influences transcriptomic responses to true microgravity aboard the ISS.

Formulation of a Spaceflight Resilience Index (SRI) for BLSS crop selection

To translate our findings into actionable breeding guidelines for lunar and Martian greenhouse cultivation, we propose a composite **Spaceflight Resilience Index (SRI)** for evaluating cereal crop candidates:
equation
SRI = w? <=ft(?_0) + w? <=ft(StarchR????????S) - w? <=ft(^2+S?????C)

equation
where ?_0 represents 30-minute gravitropic curvature velocity, StarchR???????? quantifies statolith amyloplast stability, and ^2+S????? measures hyper-activation of stress kinases (CPK28/CAS). Genotypes scoring high on the SRI (such as Koz1 and Bd21) maintain rapid root anchorage and balanced auxin transport with minimal proteotoxic stress, making them ideal genetic backgrounds for spaceflight agriculture.

Mechanistic crosstalk: Calcium mechanosensation and the LZY-RLD-PIN polarity axis

Our integrated biophysical model (Figure~5) synthesizes the molecular chain of events distinguishing 1g Earth from orbit. On Earth, amyloplast statolith sedimentation delivers LZY1 to the lower plasma membrane, recruiting RLD and polarizing PIN efflux carriers to establish the classic lateral auxin gradient[Nishimura2023,Chen2023]. In microgravity, the absence of statolith sedimentation prevents directional LZY-RLD assembly. The statocyte responds through a dual compensatory mechanism: transcriptional upregulation of BdLAZY1 and BdPIN3 accompanied by a profound mechanosensory calcium burst mediated by BdMSL10, BdCAS, and BdCPK28. Downstream, this activation alters cell wall loosening via alpha-expansins (BdEXPA1) and modulates mixed-linkage glucan deposition (BdCSLD1) within the grass Type II wall.

Monocot-specific spaceflight responses and BLSS implications

The cross-species comparison confirmed that while core stress modules (ROS detoxification, HSP chaperones, and boundary-layer gas exchange via SHMT2) are conserved across land plants (p = 3.54 x 10?^1^2, Odds Ratio = 14.16), structural adaptations diverge sharply. Cereal crops possess specialized Type II cell walls and unique tiller/root setpoint architecture regulated by DRO1 and LAZY1. Extrapolating spaceflight findings solely from dicot models (such as Arabidopsis) overlooks these cereal-specific traits. By establishing the AstroGrass resource, we provide the foundational multi-omics map required to engineer resilient cereal crops for long-duration human space exploration.

Limitations and future directions

This study has several limitations. First, while candidate association mapping identified 29 high-confidence loci, full-scale de novo GWAS across 300+ sequenced accessions will be required to resolve fine-scale QTL architecture. Second, OSD-375 examined three ecotypes; future orbital missions should evaluate multi-accession panels selected via the Spaceflight Resilience Index. Third, functional validation of candidate genes via CRISPR-Cas9 in Brachypodium and dwarf wheat lines will establish causal mechanisms.

Methods

Gravitropic reorientation assays

Gravitropic reorientation was quantified using the 2D clinostat protocol adapted from Barker et al.[Barker2016]. Seeds of Brachypodium distachyon accessions were surface sterilized in 70\% ethanol for 15 minutes, sown horizontally on 1\% agar plates containing half-strength Linsmaier-Skoog (LS) salts, and stratified at 4 deg C in darkness for 3 days. Plates were transferred to a controlled environment chamber (22 deg C, 16h light/8h dark) and positioned vertically for 4 days. On day 4, plates were scanned at 600 dpi to record baseline root positions, then rotated 90 deg\ in darkness. After defined gravitropic stimulation periods (10, 20, or 30 minutes), plates were transferred to a 2D clinostat rotating at 1 RPM for 4 hours to arrest further gravitropic curvature. Post-clinostat scans were aligned with pre-rotation images, and root tip curvature angles were measured using RootNav 2 and the CyVerse Discovery Environment[Yasrab2019]. Presentation times were calculated using the L and H models of Perbal et al.[Perbal2002].

NASA OSDR OSD-375 data acquisition

Processed RNA-Seq gene expression data (count matrices and differential expression tables) from OSDR study OSD-375[Su2023] were downloaded programmatically via the OSDR REST API (https://osdr.nasa.gov/osdr/data/osd/files/375). Sample metadata (ISA-JSON format) was parsed to map samples to ecotypes (Bd21, Bd21-3, Gaz8), tissues (root, shoot), and conditions (spaceflight vs.\ ground control). The study used APEX Growth Units in the VEGGIE facility aboard the ISS (SpaceX CRS-14, launched April 2, 2018), with seedlings grown for 5 days (24h germination + 4 days growth) and preserved in RNAlater[Su2023].

SNP and genotype data

Brachypodium distachyon variation data were obtained from multiple public sources: (1) Ensembl Plants release 60 VCF files for B.\ distachyon v3.0 (https://plants.ensembl.org/Brachypodiumd????c?y??); (2) the BrachyPan pan-genome diversity panel (54 inbred lines)~ Gordon2017; and (3) JGI Phytozome v13 (https://phytozome-next.jgi.doe.gov/). SNPs within 10 kb of gravitropism candidate genes were extracted and annotated for predicted functional effects.

Candidate gravitropism gene curation

A curated set of gravitropism candidate genes was compiled from: (a) Brachypodium orthologs of Arabidopsis genes annotated to GO:0009630 (gravitropism) and GO:0009958/GO:0009959 (positive/negative gravitropism); (b) PIN auxin efflux carriers, AUX1/LAX influx carriers, TIR1/AFB F-box receptors, ARF and Aux/IAA transcription factors; (c) LAZY/SGR gravity signaling family members; (d) statolith biogenesis genes (PGM, ADG1); (e) calcium signaling genes highlighted by Su et al.[Su2023] (CPK28, CAS, CRK28). Orthologs were identified using Ensembl Compara and reciprocal BLAST.

Differential expression analysis

Processed GeneLab count matrices from OSD-375 were analyzed using DESeq2 with the design formula ~ condition (spaceflight vs.\ ground control), performed separately for each ecotype x tissue combination. Genes with adjusted $p < 0.05$ (Benjamini-Hochberg) and |log?FC| > 0.5 were classified as differentially expressed.

GWAS--spaceflight integration

Enrichment of gravitropism candidate genes among spaceflight DEGs was tested using Fisher's exact test, with all expressed genes as background. Trait-level enrichment was assessed for gravitropism-related GO terms and pathways using clusterProfiler with custom Brachypodium GO annotations.

Cross-species comparison

Brachypodium--Arabidopsis orthologous gene pairs were obtained from Ensembl Compara (one-to-one orthologs). Spaceflight DEGs from OSD-375 were compared with our published Arabidopsis spaceflight meta-analysis (2,550 consensus genes from 17 studies[Barker2026]) using hypergeometric enrichment tests. Pathway-level comparison was performed using Gene Ontology enrichment of shared vs.\ species-specific DEG sets.

Alternative splicing analysis

Differential splicing events in OSD-375 were obtained from our multi-species alternative splicing analysis[BarkerSplicing2026], which used rMATS and SUPPA2 on aligned BAM files to identify differential exon usage, intron retention, and alternative 5'/3' splice site events between spaceflight and ground control samples.

Data and code availability

All code and processed data are available at https://github.com/dr-richard-barker/brachypodium-gwas-spaceflight. The interactive results dashboard is available at https://dr-richard-barker.github.io/brachypodium-gwas-spaceflight/. Raw sequencing data are available from NASA OSDR under accession OSD-375 (https://osdr.nasa.gov/bio/repo/data/studies/OSD-375; DOI: 10.26030/2x6b-3v89). SNP data sources are documented in the repository.

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Author contributions

R.B.\ conceived the study, designed and implemented the analysis pipeline, performed gravitropic reorientation assays, and wrote the manuscript.

Competing interests

The author declares no competing interests.

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